Q. 1. Consider the following set of requirements for a university database used to maintain student grade reports (transcripts).

A. The university records each student's name, social security number (SSN), address, phone, birthdate, major department, and degree program (B.A., B.S., etc.). An SSN uniquely designates a student.

B. Each department is described by a name, department code, office number, office phone, and college. Name and code values each uniquely characterize a department.

C. Each course has a course name, description, course number, number of quarter hours credit, and offering department. The combination of a course number and an offering department uniquely designates a course.

D. Each section has an instructor, semester, year, course and section number. The section number distinguishes different sections of the same course that are taught during the same quarter and year.

E. A grade report has a student, section, and grade (A,B,C, and D). The combination of a student and section uniquely characterizes a grade report.

Draw the E-R diagram. Create the database in MySQL. Frame and execute the SQL queries for the following:

1. List name of all students who have registered for courses in more than one department.

2. List name and SSN of all students who have registered for more than one course in a specified department.

3. List name of all students who have got at least B grade in all the courses taken by him/her.

4. List name of all students who have got A in all the courses offered by CS department.

5. List the highest grade obtained in each course by students majoring in each of the departments.

Q.2) Define the schema for the following databases with specific data type and constraints, the table name and its fields name are to be taken from database description which are given below:

A database is being constructed for storing sales information system. A product can be described with a unique product number, product name, selling price, manufacturer name. The product can sale to a particular client and each client have it own unique client number, client name, client addresses, city, pin code, state and total balance to be required to paid. Each client order to buy product from the salesman. In the order, it has unique sales order number, sales order date, client number, salesman number (unique), billed whole payment by the party or not and its delivery date. The salesman have the name, addresses, city, pin code, state, salary of the sales man, delivery date, total quantity ordered, product rate.

Write the SQL queries for the following –

a) List the various products available.
b) Find the names of all clients having 'a' as the second letter in their names.
c) List all the clients who are located in Kolhapur.
d) Find the products whose selling price is greater than 2000 and less than or equal to 5000
e) Add a new column NEW_PRICE into the product_master table.
f) Rename the column product_rate of Sales_Order_Details to new_product_rate.
g) Display the order number and date on which the clients placed their order.
h) Change the delivery date of order number ON01008 to 16-08-08
i) Change the bal_due of client_no CN01003 to 1200
j) Find the product with description as 'HDD1034' and 'DVDRW'
k) List the names, city and state of the clients not in the state of 'ASSAM'
l) Write a cursor to List the products in sorted order of their description.
m) Write a procedure/function to Delete all the records having delivery date before 25th August, 2008.

Q.3) Design the MySQL Database with following entities,

- SAILORS (SID: INTEGER, SNAME:STRING, RATING:INTEGER(Must be in between 1 to 10), AGE:REAL)
- BOATS (BID: INTEGER, BNAME: STRING, COLOR: STRING)
- RESERVES (SID: INTEGER, BID: INTEGER, DAY: DATE)

Make appropriate tables and add required data for the above database. **Frame and execute the SQL queries for the following:**

a) Find the names of sailors who have reserved a red or a green boat.

b) Find the names of sailor's who have reserved both a red and a green boat.

c) Find the sids of all sailor's who have reserved red boats but not green boats.

d) Find sailors whose rating is better than some sailor called Rajesh.

e) Find the sailor's with the highest rating using ALL.

f) Find the name and age of the oldest sailor.

g) Find the names of sailors who are older than the oldest sailor with a rating of 10.

h) Displays all the sailors according to rating (Lower Rating First), if rating is same then sort according to age (Younger First).

i) Find the age of youngest sailor with age >= 18 for each rating with at least 2 such sailors.

j) Write a procedure to delete records from sailors table by reading SID from Keyboard.

Q.4) Design the MySQL Database with following entities,

- SAILORS (SID: INTEGER, SNAME:STRING, RATING:INTEGER(Must be in between 1 to 10), AGE:REAL)
- BOATS (BID: INTEGER, BNAME: STRING, COLOR: STRING)
- RESERVES (SID: INTEGER, BID: INTEGER, DAY: DATE)

Make appropriate tables and add required data for the above database. **Frame and execute the SQL queries for the following:**

1. Find the names of sailors who have reserved boat number 123.
2. Find names of the sailors who have reserved at least one boat.
3. Create a view for Expert Sailors (A sailor is a Expert Sailor if his rating is more than 7).
4. Find average age of Expert sailors.
5. Write the following queries on Expert Sailor View.
   5.1 Find the Sailors with age > 25 and rating equal to 10.
   5.2 Find the total number of Sailors in Expert Sailor view.
   5.3 Find the number of Sailors at each rating level (8, 9, 10).
   5.4 Find the age of Oldest as well as Youngest Expert Sailor.
6. Write cursor to display sailor records according to rating.
7. Write appropriate cursor to update rating of sailors by 2 if rating is less than 5, by 1 if rating is >5 and doesn’t change the rating if it is equal to 10.

Q.5) Design the MySQL Database with following entities,

Customer (**Cust id : integer,** cust_name: string)

Item (**item_id: integer,** item_name: string, price: integer)

Sale (**bill_no: integer,** bill_data: date, **cust_id: integer, item_id: integer**, qty_sold: integer)

For the above schema, perform the following—

a) Create the tables with the appropriate integrity constraints.
b) Insert around 10 records in each of the tables
c) List all the bills for the current date with the customer names and item numbers
d) List the total Bill details with the quantity sold, price of the item and the final amount.
e) List the details of the customer who have bought a product which has a price>200.
f) Give a count of how many products have been bought by each customer
g) List the item details and count which are sold as of today.
h) Write a procedure to Give a list of products bought by a customer having cust_id as 5
i) Create a view which lists out the bill_no, bill_date, cust_id, item_id, price, qty_sold, amount.
j) Create a view which lists the daily sales date wise for the last one week.

Q.6) Design the MySQL Database Schema for Student Library scenario

Student (**Stud_no : integer,** Stud_name: string)

Membership (**Mem_no: integer, Stud_no: integer**)

Book (**book_no: integer**, book_name:string, author: string)

Iss_rec **(iss_no:integer**, iss_date: date, **Mem_no: integer**, **book_no: integer**)

For the above schema, perform the following—

a) Create the tables with the appropriate integrity constraints
b) Insert around 10 records in each of the tables
c) List all the student names with their membership numbers
d) List all the issues for the current date with student and Book names
e) List the details of students who borrowed book whose author is CJDATE
f) Give a count of how many books have been bought by each student
g) Give a list of books taken by student with stud_no as 5
h) List the book details which are issued as of today
i) Create a view which lists out the iss_no, iss _date, stud_name, book name
j) Create a view which lists the daily issues-date wise for the last one week

Q.7) Design the MySQL Database Schema for a Employee-pay scenario

Employee (**emp_id : integer**, emp_name: string)
Department (**dept_id: integer**, dept_name:string)
Paydetails (**emp_id : integer**, **dept_id: integer**, basic: integer, deductions: integer, additions: integer, DOJ: date)
Payroll **(emp_id : integer**, pay_date: date)

For the above schema, perform the following—

a) Create the tables with the appropriate integrity constraints
b) Insert around 10 records in each of the tables
c) List the employee details department wise
d) List all the employee names who joined after particular date
e) List the details of employees whose basic salary is between 10,000 and 20,000
f) Give a count of how many employees are working in each department
g) Give a names of the employees whose netsalary>10,000
h) Write a cursor to List the paydetails for all employee.
i) Create a view which lists out the emp_name, department, basic, dedeuctions, netsalary
j) Create a view which lists the emp_name and his netsalary

Q.8) Design the MySQL Database Schema for Video Library scenario

Customer (cust_no: integer,cust_name: string)

Membership (**<u>Mem_no: integer</u>**, **cust_no: integer**)

Cassette (**<u>cass_no:integer</u>**, cass_name:string, Language: String)

Iss_rec **(iss_no: integer**, iss_date: date, **mem_no: integer**, **cass_no: integer**)

For the above schema, perform the following:-

a) Create the tables with the appropriate integrity constraints
b) Insert around 10 records in each of the tables
c) List all the customer names with their membership numbers
d) List all the issues for the current date with the customer names and cassette names
e) List the details of the customer who has borrowed the cassette whose title is " The Legend"
f) Give a count of how many cassettes have been borrowed by each customer
g) Give a list of book which has been taken by the student with mem_no as 5
h) Write a procedure to List the cassettes issues for today.
i) Write a trigger to delete a Customer record.
j) Create a view which lists outs the iss_no, iss_date, cust_name, cass_name
k) Create a view which lists issues-date wise for the last one week

Q.9) Design the MySQL Database Schema for student-Lab scenario

Student (**stud_no: integer**, stud_name: string, **class: string**)

Class (**<u>class: string,</u> descrip: string**)

Lab (**<u>machi_no: integer</u>**, Lab_no: integer, description: String)

Allotment (**Stud_no: Integer, mach_no: integer, dayof week: string)**

For the above schema, perform the following—

a) Create the tables with the appropriate integrity constraints
b) Insert around 10 records in each of the tables
c) List all the machine allotments with the student names, lab and machine numbers
d) List the total number of lab allotments day wise
e) Give a count of how many machines have been allocated to the 'CSIT' class
f) Give a machine allotment etails of the stud_no 5 with his personal and class details
g) Count for how many machines have been allocatedin **Lab_no 1** for the day of the week as "Monday"
h) How many students class wise have allocated machines in the labs
i) Create a view which lists out the stud_no, stud_name, mach_no, lab_no, dayofweek
j) Create a view which lists the machine allotment details for "Thursday".

Q.10) ROADWAY TRAVELS "Roadway Travels" is in business since 1977 with several buses connecting different places in India. Its main office is in Pune. The company wants to computerize its operations in the following areas:

- Reservations
- Ticketing
- Cancellations

Reservations: Reservations are directly handled by booking office. Reservations can be made 60 days in advance in either cash or credit. In case the ticket is not available, a wait listed ticket is issued to the customer. This ticket is confirmed against the cancellation.

Cancellation and modification: Cancellations are also directly handed at the booking office. Cancellation charges will be charged. Wait listed tickets that do not get confirmed are fully refunded.

Analyze the problem and following with the entities in it. Identify what Data must be persisted in the databases.

1. Bus: - Bus No, Source, Destination, Couch Type

2. Reservation: - PNR NO, Journey date, Contact No, No-of-Seats, Bus No, Address

3. Ticket: - Ticket No, Journey date, Age, Dep-Time, Sex, Source, Destination, Bus No

4. Passenger: - Ticket No, PNR NO, Age, Sex, Contact NO, Name

5. Cancellation: -PNR NO, Journey date, Seat No, Contact No

Design the suitable ER Diagram and execute following queries.

a) Create the tables with the appropriate integrity constraints
b) Insert around 10 records in each of the tables.
c) Display all the names of male passengers.
d) Find the ticket numbers of the passengers whose name start with 'r' and ends with 'h'.
e) Find the names of passengers whose age is between 30 and 45.
f) Display the sorted list of passenger's names.
g) Find the number of tickets booked by a passenger where the number of seats is greater than 1.
h) Display the Total number of Reservation per bus number.
i) Write a cursor to display the ticket numbers and names of all the passengers.

Q.11) A student is described by a unique Roll Number, Name Adress, and Semester. Each student enrolls himself in an Academic programme offered by a Department. Academic programmes have programme name(unique), duration, a programme code(unique) and a list of courses (both core and elective course) while the departments have department code (unique), department name (unique), HoD who is a Teacher and list of courses offered by it. Each teacher is described by employee code (unique), name, department and designation. A student registers some courses in a semester. A course is described by a unique course number, title of the course, credit allotted for the course and offering department. Database stores the grades obtained by different student in different courses registered by him/her in different semesters. Database also stores information about the courses offered by a department in a semester, the corresponding teacher(s) for each course.

Write the SQL queries for the following –

a) List the available courses from the CSE department.
b) List the all students in a particular semester.
c) List the students who earned CGPA greater than or equal to 8.5
d) How much subjects are registered by a student in each semester.
e) List the common students who are allotted the same courses of both the programme MCA and M.Tech.
f) List the total number of student enrolled in the subject DBMS.
g) Retrieve the semester of the student under DBMS subject.
h) Retrieve all the student name and arrange into ascending order.
i) Modify a student address Kolhapur to Pune where sdt_id='CSI08002'.
j) Find the total credit point of student required to complete for a course like MCA.
k) Retrieve all the students located at 'Pune'.
l) Find the total number of department in our database.
m) Write a Cursor to List all the courses which are related to computer science.
n) Write a trigger to delete all corresponding records from assigned table if student deleted.

Q. 12: Bank Database:

A bank database keeps record of the details of customers, accounts, loans and transactions such as deposits or withdraws. Customer record should include customer id, customer name, address, age, contact number, email id etc., accounts details involves account number, account type(fixed account, savings account, monthly account etc), date of creation of the account, balance. Transaction detail keeps information about amount deposited or withdrawn to/from a particular account and the date of transaction. The database should also store record of loans which include loan amount, loan date and the account number to which the loan is granted.

Make appropriate tables for the above database and try to find out the following queries: (Apply appropriate triggers whenever required)

a) List the details of account holders who have a 'savings' account.
b) List the Name and address of account holders with loan amount more than 50,000.
c) Change the name of the customer to 'ABC' whose account number is 'TU001'
d) List the account number with total deposit more than 80,000.
e) List the number of fixed deposit accounts in the bank.
f) Display the details of customers who created their accounts between '20-jan-08' to '20-aug-08'.
g) Display the detailed transactions on 28th Aug, 2008.
h) Display the total amount deposited and withdrawn on 29th Aug, 2008.
i) List the details of customers who have a loan.
j) Write a cursors to display Savings and Loan information of all customers.
k) Write a trigger to delete the customer.

Q.13) Hospital information system:

Patients - indoor/outdoor, medicines/lab tests (including results) prescribed to patients, information if a patient if referred to other expert/hospital. Doctors - specialization, patients attended etc. Different wards/beds and patients alloted to them etc.

Patient registration form should include Registration number, Patient name, Address, Gender, Bed number, date of registration, refer doctor id etc.

Doctor information should include Doctor code, Doctor Name, Specialization etc.

Lab test information should include Test name, test number, test date, results and referred doctor's code.

Bed information should include bed number, ward number and status (whether allotted or not).

Write suitable Queries for following statements:

1. Display the details of patients admitted between '20-jul-02' and '20-aug-08'.
2. Change the name of the patient to 'Ram' whose patient id='PT011'
3. Display the names of the patients and lab test results performed on '20-jul-08'.
4. Display the number of patients taking treatment under doctor ='ABC'.
5. Retrieve the name of doctor who is taking care of maximum number of patients.
6. Change the bed number of the patient to 456 where patient id='PT023'
7. Change the status of bed with bed number 123 with 'not allotted'.
8. List the bed details which are free in ward number 10.
9. List the name of male patients in ward no 13 taking treatment under doctor 'XYZ'
10. List the details of patients with age more than 50 taking treatment under a doctor, whose name like 'das'.
11. Write a cursor to display details of all patients from allocated doctors to allocated beds.

Q.14) Consider the following relational schema. An employee can work in more than one department.

Emp (*eid:* integer, *ename:* string, *salary:* real)

Works (*eid:* integer, *did:* integer)

Dept (*did:* integer, *dname:* string, *managerid:* integer, *floornum:* integer)

Write the following Queries:

1. Print the names of all employees who work on the 10th floor and make less than Rs. 50,000.
2. Print the names of all managers who manage three or more departments on the same floor.
3. Write a procedure to Give every employee who works in the toy department a 10 percent raise.
4. Print the names and salaries of employees who work in both the toy department and the Music department.
5. Print the names of employees who earn a salary that is either less than Rs. 10,000 or more than Rs. 100,000.
6. Print all of the attributes for employees who work in some department that employee Abhishek also works in.
7. Print the names of employees who make more than Rs. 20,000 and work in either the video department or the toy department.
8. Print the name of each employee who earns more than the manager of the department that he or she works in.
9. Print the name of each department that has a manager whose last name is pantamabe and who is neither the highest-paid nor the lowest-paid employee in the department.
10. Write a Procedure to Print the names of all employees who work on the floor(s) where Amar Arora works.
11. Write a trigger to update department id from department, corresponding table must be reflected accordingly.

Q15. Create Order Management System using MongoDB and Implement Following Statements

1. Display the name of customers who have maximum orders.
2. Display the Mob No of customers who have highest Buying Total.
3. Display how many customers are there in customer collection.
4. Using collection of customer, and $exists, tell me how many customers belongs from pune city.
5. Find the customer who purchased shoes and cloth product.
6. Find the top 10 buyers.
7. Display all the orders where total amount is >1000.
8. Display All the customers with corresponding buying price.
9. write a MapReduce function which will return the Total Price per Customer.

Q.16) Create Employee Project Management System using MongoDB.

Dept have dept-no, dname, LOC fields
emp have emp-no, ename, designation,sal attributes
project have proj-no, proj-name, status attributes

Implement following queries

i) List all employees of 'INVENTORY' department of 'PUNE' location.
ii) Give the names of employees who are working on 'Blood Bank' project.
iii) Give the name of managers from 'MARKETING' department.
iv) Give all the employees working under status 'INCOMPLETE' projects

Q.16 Create **restaurants Management System** using MongoDB with following document fields

```
{
 "address": {
   "building": "1007",     "coord": [ -73.856077, 40.848447],
   "street": "Tilak Road",    "zipcode": "411046"
 },
 "area": "Pune",   "cuisine": "Bakery",
 "grades": [
{ "date": { "$date": 1393804800000 }, "grade": "A", "score": 2 },
{ "date": { "$date": 1378857600000 }, "grade": "A", "score": 6 },
{ "date": { "$date": 1358985600000 }, "grade": "A", "score": 10 },
{ "date": { "$date": 1322006400000 }, "grade": "A", "score": 9 },
{ "date": { "$date": 1299715200000 }, "grade": "B", "score": 14 }
 ],
 "name": "Jai Bake Shop",   "restaurant_id": "30075445"
}
```

Implement Following Statements

1. Write a MongoDB query to display all the documents in the collection restaurants
2. Write a MongoDB query to display the fields restaurant_id, name, area and cuisine for all the documents in the collection restaurant.
3. Write a MongoDB query to display the fields restaurant_id, name, area and cuisine, but exclude the field _id for all the documents in the collection restaurant.
4. Write a MongoDB query to display the fields restaurant_id, name, area and zip code, but exclude the field _id for all the documents in the collection restaurant.
5. Write a MongoDB query to display all the restaurant which is in the borough Bronx.
6. Write a MongoDB query to display the first 5 restaurant which is in the area pune.
7. Write a MongoDB query to display the next 5 restaurants after skipping first 5 which are in the area pune.
8. Write a MongoDB query to find the restaurants who achieved a score more than 90.
9. Write a MongoDB query to find the restaurants that achieved a score, more than 80 but less than 100.
10. Write a MongoDB query which will select all documents in the restaurants collection where the coord field value is Double.
11. Write a MongoDB query which will select the restaurant Id, name and grades for those restaurants which returns 0 as a remainder after dividing the score by 7.
12. write a MapReduce function which will return the Total rating score per restaurant.

Q.17) Consider following and design MongoDB Database

Suppliers (*sid:* integer, *sname:* string, *city:* string)

Parts (*pid:* integer, *pname:* string, *color:* string)

Orders (*sid:* integer, *pid:* integer, *quantity:* integer)

Implement following:

1. For each supplier from whom all of the following things have been ordered in quantities of at least 150, print the name and city of the supplier: a blue gear, a red crankshaft, and a yellow bumper.
2. Print the names of the purple parts that have been ordered from suppliers located in Mumbai, Pune, or Kolhapur.
3. Print the names and cities of suppliers who have an order for more than 150 units of a yellow or purple part.
4. Print the *pid*s of parts that have been ordered from a supplier named Amer but have also been ordered from some supplier with a different name in a quantity that is greater than the Amer order by at least 100 units.
5. Print the names of the suppliers located in Kolhapur.
6. Print all available information about suppliers that supply green parts.
7. For each order of a red part, print the quantity and the name of the part.
8. Print the names of the parts that come in both blue and green. (Assume that no two distinct parts can have the same name and color.)
9. Print (in ascending order alphabetically) the names of parts supplied both by a Madison supplier and by a Berkeley supplier.
10. Print the names of parts supplied by a Mumbaikar supplier, but not supplied by any Punekar supplier. Could there be any duplicates in the answer?
11. Print the total number of orders.
12. Print the largest quantity per order for each *sid* such that the minimum quantity per order for that supplier is greater than 100.
13. Write a MapReduce to Display the total number of parts per supplier.

Q.18) Consider the following relations about music:

Artist: Name, Type, Country

where Name is the primary key and the value of attribute Type is either 'PERSON' or 'BAND'.

Album: Title, Artist, Year, Type, Rating

where (Title, Artist) is the primary key, Artist is a foreign key referencing relation Artist, the value of Type is either 'STUDIO', 'LIVE' or 'COMPILATION', and Rating is an integer from 1 to 5.

Track list: Album Title, Album Artist, Track No, Track Title, Track Length

where (Album Title, Album Artist, Track No) is the primary key and (Album Title, Album Artist) is a foreign key referencing relation Album

Create a database for this schema and populate it. Tables Artist and Album must each contain at least 5 tuples, and each album must consist of at least 5 tracks in Track list. Do not use NULL.

Write the following queries.

(1) Display the artists who released a live album and a compilation in the same year.

(2) Display the artists who have only released studio albums.

(3) List albums which have a higher rating than every previous album by the same band.

(4) List live albums released by British artists and having a higher rating than the average rating of all albums released in the same year.

(5) List songs shorter than 2 minutes and 34 seconds from albums rated 4 or 5 stars and released in the last 20 years. Include the album title and artist name in the output.

(6) Find the average total running time of all albums released in the '90s and having at least 10 tracks. (Assume no track is missing in the track list)

(7) List artists who have never released two consecutive studio albums more than 4 years apart.

(8) List artists who have released more live and compilation albums (together) than studio ones.

(9) Among artists who have released at least 3 studio albums, 2 live albums and one compilation, list those whose every album is rated no less than 3.

(10) List artists who released no fewer studio albums than any other artist from the same country.

(11) List pairs of albums released by artists of different countries in the same year and such that one has a higher rating than the other.

(12) Sort albums by the ratio (highest first) between their rating and the number of tracks they consist of. (Assume no track is missing in the track list)

Q.19) Our toy-store database has the following schema:

Product (pid:integer, name: varchar(20), min_age: integer)
Manufacturer (mid:integer, name: varchar(20), address: varchar(50))
Supplier (sid:integer, name: varchar(20), address: varchar(50))
Inventory (pid:integer, stock: integer)
Manufactures (mid:integer, pid: integer)
Supplies (sid: integer, pid: integer)

The Product relation contains information about all the toys sold by the store. The Product.pid column is the primary key for the relation. Product.name is the name of each toy, and Product.min_age indicates the minimum age recommended to play with the toy.

Manufacturer and Supplier list the names and addresses of all toy manufacturers and suppliers respectively. Manufacturer.mid and Supplier.sid are the primary keys for these relations.

Inventory indicates the number of toys in stock. Inventory.pid is a foreign key referring to Product.pid. Inventory.stock indicates the number of toys of the given type available in stock.

Manufactures and Suppliers associate products with their manufacturers and suppliers respectively. Note that there can be only one manufacturer for a product, but there can be many suppliers.

Write SQL queries for the following:

1) List the ids and names of all products whose inventory is below 5.

2) List the ids and names of all suppliers for products manufactured by "manufacturer_2". The id and name of each supplier should appear only once.

3) List the ids, names, and number in stock of all products in inventory. Order the list by *decreasing* number in stock and *decreasing* product ids.

4) List the ids and names of all products for whom there is only one supplier.

5) Find the ids and names of the products with the lowest inventory. Do NOT assume these are always products with an inventory of zero.

6) List the id and name of each supplier along with the total number of products it supplies.

7) Find the id and name of the manufacturer who produces toys on average for the youngest children.

8) Create a view that shows only toys in the Product table for children over age 6

9) Create a view that shows the id and names of toys along with the ids and names of their manufacturers.

10) Write a cursor to display Average age of each products per Manufacturer

Q.20) Shops sell items at varying prices. Customers buy items from shops. This is described by the following relations:

Shops (ShopId, name, address)

Items (ItemId, name, description)

Sells (ShopId, ItemId, price)

Customers (CustomerId, name, address)

Sales (SaleId, CustomerId, ItemId, ShopId, date)

a) Draw an E/R diagram that describes the database.

b) Create the table Sells. Invent suitable types for the attributes (itemId shall have the type char(10)) and indicate reasonable integrity constraints.

c) Print the name and address of all customers who haven't bought any item.

d) Print the number of shops that sell items with id's starting with 'EF'.

e) Print the name and address of the shop(s) that sell the item with id = 'EF123-A' at the lowest price.

f) For all customers that have bought at least one item: print the customer id and the total sum of all purchases.

g) Display total price per customer.

h) Display the name of Highest buyer.

i) Write a procedure to display total items per customer.

j) Write a trigger to delete item, corresponding tables must have reflected according.

Q.21) Create a relational database that contains the following tables and insert the following data into these tables.

STUD_MEMBER

Roll_No	*FName*	*MName*	*SName*	*Dept_ID*	*Semester*	*Contact_No*	*Gender*
1	Ankur	Samir	Kahar	1	1	272121	M
2	Dhaval	Dhiren	Joshi	1	1	232122	M
3	Ankita	Biren	Shah	1	1	112121	F
10	Komal	Maheshkumar	Pandya	2	3	123123	F
13	Amit	Jitenkumar	Mehta	3	3	453667	M
23	Jinal	Ashish	Gandhi	2	1	323232	M
22	Ganesh	Asha	Patel	2	3	124244	M
4	Shweta	Mihir	Patel	3	1	646342	F
7	Pooja	Mayank	Desai	3	3	328656	F
8	Komal	Krishnaraj	Bhatia	2	3	257422	F
43	Kiran	Viraj	Shah	1	1	754124	F

DEPARTMENT

Dept_ID	*Dept_Name*
1	Information Technology
2	Electrical
3	Civil
4	Mechanical
5	Chemical

Solved the Following SQL Queries:

1. Display the first name and contact number of all students.
2. Display Roll no, First name, dept id and Dept name.
3. Display the First name, Surname , contact no and Dept name whose first name contain 'a' character at any place but not studying in Civil or Mechanical department .
4. Give the name and roll no of all students of information technology who are male.
5. Display the name of departments of students who are in 1^{st} semester.
6. Display name of departments for which no students are in 3^{rd} semester.
7. Count the Total no of Female Students in each department.
8. Count the Total no of Female and Male Students and display the maximum output.
9. Count the Total no of Students in each department and display only those department which having more than 3 students.
10. Count the Total no of Students in each department arrange that calculate students in descending order.
11. Display data of student firstname members in alphabetical order.
12. Count the Total no of Students semester wise.
13. Display data of student who are currently in semester '1' but not a gender female.
14. Write a cursor to Count the Total no of Students in each department.
15. Write a procedure to display total no of students in corresponding dept (Dept name should be taken from user)

Q.22) Convert the following relation schema into MongoDB Database

Employee (empid, ename, city, salary, post, dept_no, join_date**)**

Department (dept_no, dname, location**)**

Skill (empid, skill**)** skill**-** writing, speaking, typing

Implement following statements.

1. List the employee details that are from Baroda or Ahmedabad and working in sales department.
2. Display the information of employees having typing skill and working in computer department who are currently joined.
3. List of the empid, ename, department number and skill of employee whose join date is 20^{th} of any month.
4. Calculate total experience of employee. Consider the today's date.
5. Calculate department wise total salary and display only those departments which pay maximum salary.
6. List the name of employee whose name staring with 's' or 'm' character who are working in computer department and having speaking skill.
7. Count the no of employee working in system department of Pune Location.
8. Arrange the employee name in alphabetic order whose salary between 4000 to 12000 and working in sales department.
9. Count total no of employee in Sales department.
10. Count the total no employee date wise and display only those date which having at least 2 employee.
11. Write a MapReduce operation to count total number of employee in each department.

Q.23) Consider following relational schema

TABLE : **DEPT**

DEPTNO	DNAME	LOC
10	ACCOUNTING	NEW YORK
20	RESEARCH	DALLAS
30	SALES	CHICAGO
40	OPERATIONS	BOSTON

--TABLE: **EMP**

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	SMITH	CLERK	7902	17-Dec-80	800		20
7499	ALLEN	SALESMAN	7698	20-Feb-81	1600	300	30
7521	WARD	SALESMAN	7698	22-Feb-81	1250	500	30
7566	JONES	MANAGER	7839	2-Apr-81	2975		20
7654	MARTIN	SALESMAN	7698	28-Sep-81	1250	1400	30
7698	BLAKE	MANAGER	7839	1-May-81	2850		30
7782	CLARK	MANAGER	7839	9-Jun-81	2450		10
7788	SCOTT	ANALYST	7566	9-Dec-82	3000		20
7839	KING	PRESIDENT		17-Nov-81	5000		10
7844	TURNER	SALESMAN	7698	8-Sep-81	1500	0	30
7876	ADAMS	CLERK	7788	12-Jan-83	1100		20
7900	JAMES	CLERK	7698	3-Dec-81	950		30
7902	FORD	ANALYST	7566	3-Dec-81	3000		20
7934	MILLER	CLERK	7782	23-Jan-82	1300		10

Solve following queries

1) Display all the ANALYSTs whose name doesn't ends with 'S'
2) Display all the employees who are earning more than all the managers.
3) Display all the managers working in 20 & 30 department.
4) Display all the managers who don't have a manager
5) Display the second maximum salary.
6) Display the departments that are having more than 3 employees under it.
7) Display job-wise average salaries for the employees whose employee number is not from 7788 to 7790.
8) Display department-wise total salaries for all the Managers and Analysts, only if the average salaries for the same is greater than or equal to 3000.
9) Display the third maximum salary.
10) Display all the managers & clerk who work in Accounting and Marketing (SALES) departments.
11) Display all the managers & clerks who work in Accounts and Marketing departments.
12) Display all the employees who have joined before their managers.
13) Select all the employees name along with their manager names, and if an employee does not have a manager, display him as "CEO".
14) Write a trigger to delete department, dependent table must be reflected accordingly.
15) Write a cursor to Display job-wise maximum salary.

Q.24) Consider following relational schema

TABLE : **DEPT**

DEPTNO	DNAME	LOC
10	ACCOUNTING	NEW YORK
20	RESEARCH	DALLAS
30	SALES	CHICAGO
40	OPERATIONS	BOSTON

--TABLE: **EMP**

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	SMITH	CLERK	7902	17-Dec-80	800		20
7499	ALLEN	SALESMAN	7698	20-Feb-81	1600	300	30
7521	WARD	SALESMAN	7698	22-Feb-81	1250	500	30
7566	JONES	MANAGER	7839	2-Apr-81	2975		20
7654	MARTIN	SALESMAN	7698	28-Sep-81	1250	1400	30
7698	BLAKE	MANAGER	7839	1-May-81	2850		30
7782	CLARK	MANAGER	7839	9-Jun-81	2450		10
7788	SCOTT	ANALYST	7566	9-Dec-82	3000		20
7839	KING	PRESIDENT		17-Nov-81	5000		10
7844	TURNER	SALESMAN	7698	8-Sep-81	1500	0	30
7876	ADAMS	CLERK	7788	12-Jan-83	1100		20
7900	JAMES	CLERK	7698	3-Dec-81	950		30
7902	FORD	ANALYST	7566	3-Dec-81	3000		20
7934	MILLER	CLERK	7782	23-Jan-82	1300		10

Solve following queries

1) Display all the employees whose naming is having letter 'E' as the last but one character
2) Display all the employees who total salary is more than 2000.
3) Display name of all the employee who joined after 30th Feb 81 with department name.
4) Display all the employees who are getting some commission in department 20 & 30.
5) Select all the employees who work in DALLAS.
6) Select employee number, job & salaries of all the Analysts who are earning more than any of the managers.
7) Select department name & location of all the employees working for CLARK.
8) Display all the salesmen who are not located at DALLAS.
9) Display all the departmental information for all the existing employees and if a department has no employees display it as "No employees".
10) Get all the matching & non-matching records from both the tables.
11) List all the employees whose salaries are greater than their respective departmental average salary.
12) Write a trigger to delete department, dependent table must be reflected accordingly.
13) Write a cursor to Display Department-wise maximum salary.

Q.25)

Create LOCATION Table with columns Location_Id, Regional_Group. Constraints on LOCATION table: Location_Id Primary Key.

Create DEPARTMENT Table with columns Department_Id, Name, Location_ID. Constraints on DEPARTMENT table: Department_Id Primary Key, Location_Id references LOCATION table.

Create JOB Table with columns Job_Id, Funcation.

Create EMPLOYEE Table with columns Employee_Id, Last_Name, First_Name, Middle_Name, Job_Id, Manager_Id, Hire_Date, Salary, Comm, Department_ID. Constraints on EMPLOYEE table: Employee_Id Primary Key, Last_Name NotNull, Department_Id references DEPARTMENT table.

Design above relational schema into MongoDB Database. And Solve following queries.

1. List all job details.
2. List all the locations.
3. List out the employees who are working in department 10 or 20.
4. List out the employee id, last name in ascending order based on the employee id.
5. List out the employee id, name in descending order based on salary column.
6. List out the employee details according to their last name in ascending order and salaries in descending order.
7. List out the department wise maximum salary, minimum salary, average salary of the employees
8. How many employees in Hired January month.
9. List out the no. of employees for each month and year, in the ascending order based on the year, month.
10. List out the employee's annual salary with their names only.
11. Which is the department id, having greater than or equal to 3 employees joined in April 1985.
12. Write a MapReduce function to display average working age of employees per department.

Q.26) Create LOCATION Table with columns Location_Id, Regional_Group. Constraints on LOCATION table: Location_Id Primary Key.

Create DEPARTMENT Table with columns Department_Id, Name, Location_ID. Constraints on DEPARTMENT table: Department_Id Primary Key, Location_Id references LOCATION table.

Create JOB Table with columns Job_Id, Funcation.

Create EMPLOYEE Table with columns Employee_Id, Last_Name, First_Name, Middle_Name, Job_Id, Manager_Id, Hire_Date, Salary, Comm, Department_ID. Constraints on EMPLOYEE table: Employee_Id Primary Key, Last_Name NotNull, Department_Id references DEPARTMENT table.

Design above relational schema into MongoDB Database. And Solve following queries.

1. Display the employee who got the maximum salary.
2. List out the common jobs in Research and Accounting Departments in ascending order.
3. List out first name,last_name,salary, commission for all employees.
4. List out employee_id,last_name,department_id for all employees and rename employee_id as “ID of the employee”, last_name as “Name of the employee”, department_id as “department ID”.
5. List out the employees who are earning salary between 3000 and 4500.
6. List out the department id having at least four employees.
7. Find out no. of employees working in “Sales” department.
8. How many employees who are working in different departments and display with department name.
9. List our employees with their department names.
10. Display employees with their designations (jobs).
11. Display the employees who are working in Sales department.
12. Display the employees who are working as “Clerk”.
13. Display employee details with all departments.
14. How many jobs in the organization with designations.
15. Write a MapReduce function to display total number of employees per department.

Q.27) Create the table DISTRIBUTOR with the fields (DNO, DNAME, DADDRESS, DPHONE)

Constraints on table DISTRIBUTOR: dno primary key, dname not null.

Create the table ITEM1 with the fields (ITEMNO, ITEMNAME, COLOR, WEIGHT) Constraints on table ITEM1: itemno primary key, itemname not null, check for weight>0

Create the table DIST_ITEM with the fields (DNO, ITEMNO, QTY): Constraints of table DIST_ITEM: dno references DISTRIBUTOR table, itemno references ITEM table.

1. Create a view LONDON_DIST on DIST_ITEM which contains only those records where distributors are from London. Make sure that this condition is checked for every DML against this view.
2. Display detail of all those item that have never been supplied.
3. Count the number of items having the same color but not having weight between 20 and 100
4. Display all those distributors who have supplied more than 1000 parts of the same type.
5. Display the average weight of items of same colour provided at least one items have that colour.
6. Display the position where a distributor name has an 'OH' in its spelling somewhere after the forth character.
7. Count the number of distributors who have a phone connection and are supplying item number 'I100
8. Create a view on the table in such a way that the view contains the distributor name, item name and the quantity supplied.
9. List the name, address and phone number of distributors who have the same three digits in their number as 'Mr. Talkative'.
10. List all distributor names who supply either item I01 or I07 and the quantity supplied is more than 100.
11. Display the data of the top three heaviest ITEMS.
12. Count the total quantity group by itemno.
13. Delete all those items that have been sulpplied only once.
14. List the names of distributors who have an 'A' and also a 'B' somewhere in their names.

Q.26)

Create the table WORKER with the fields (worker_id, name, wage_per_hour, specialized_in, manager_id) Constraints on table WORKER: worker_id primary key, name not null, manager_id primary key, check for wage_per_hour>=0.

Create the table JOB with the fields (job_id, type_of_job, status):

Create the table JOB_ASSIGNED with the fields (worker_id, job_id, starting_date, number_of_days)

Constraints on table JOB_ASSIGNED: worker_id references WORKER table, job_id references JOB table.

1. Display the date on which each worker is going to end his presently assigned job.
2. Display how many days remain for each worker to finish his job.
3. Display the STARTING_DATE in the following format - 'The fifth day of month of October, 2004'.
4. Change the status to 'Complete' for all those jobs, which started in year 2008.
5. What is total number of days allocated for printing on the goods for all the workers together.
6. Which workers receive higher than average wage per hour.
7. Display details of workers who are working on more than one job.
8. Which workers having specialization in polishing start their job in September?
9. Display details of workers who are specialized in the same field as that of Mr.Cacophonix or have a wage per hour more than any of the workers.
10. Display job details of all those jobs where at least 25 workers are working.
11. Display all those jobs that are already incompleted.
12. Find all the jobs, which begin within the next two weeks.
13. List all workers who have their wage per hour ten times greater than the wage of their managers.
14. List the names of workers who have been assigned the job of Packing.
15. Find the names of the workers who are getting more then 50 Rs. as wages per hour.
16. Find the jobs which are assigned after 31-DEC-2008.

Q.28) Create the following table named table as CUSTOMER with following fields- Cust_No, First_Name, Last_Name, Address, City, State, Pin, B_Date, Status. Constraints on table CUSTOMER: Cust_No Primary Key, First_Name Not Null and the values for status must be in (‘V’,’I’,’A’).

Convert the above relation schema in MongoDB and Implement following queries

1. Display all the records from the table where state is KARNATAKA.
2. Delete the row from the table where PIN CODE is 576201.
3. Change the ADDRESS as “KAVI MUDDANNA MARG” AND PIN=576104 where CUST_NO=1003.
4. Delete the records of KARNATAKA state from the table and then retrieve all the records back.
5. Select all the records with single occurrence of state from the table.
6. Sort and display the customer data, in the alphabetic order of city.
7. Sort and display the state field in the in descending order.
8. Retrieve records of Karnataka / Kerala customers who are ACTIVE (‘A’).
9. Retrieve rows where name contains the word RAJ embedded it.
10. Display all the rows whose dates are in the range of 10-JAN-70 and 31-JUL-96.
11. Write a mapreduce function to calculate total customer per City